$$p_k = \frac{e^{s_k}}{\sum_{j=1}^{K} e^{s_j}}, \qquad \mathcal{L} = -\log p_c$$

$$p_c = 0.2: \ \mathcal{L} \approx 1.609; \quad p_c = 0.8: \ \mathcal{L} \approx 0.223$$